

Juan Fernando Garcia Navas

• Open to internships · July 2027



ENSIC Degree Double degree · ENSIC + Master + UIS

Programme Chemical and process engineering · ENSIC 3rd year + Master

Specialisation Processes for Energy and the Environment

Languages English C1 French B2 Spanish native

Experience

- 2026
Ammonia unit · Technip Energies
Full design of a 2 700 t/day unit, defended before their engineers.
- 2025 · 2026
EscapeScience · Collège Grandville
Eight scientific challenges for 120 pupils, with our own budget and calendar.
- 2025
1st prize · Poster
Colombian National Symposium on Catalysis.
- 2025
Cooling tower · UIS
From technology study to live start-up, 73.1% thermal efficiency.
- 2025
Research intern · CICAT, UIS
Synthesis and characterisation of nickel catalysts.



Publications and posters

Poster · $\text{Ni/TiO}_2/\text{Al}_2\text{O}_3$ catalysts for CO_2 methanation
Colombian National Symposium on Catalysis, 2025 · 1st prize.

SCAN

The full portfolio



juanfernandogarcia.github.io/portfolio

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Areas of interest

CO₂ capture and valorisation

Hydrogen and synthetic fuels

Process design and simulation

Energy and process optimisation

Oil and gas

Base chemicals and polymers

Process intensification

Profile

Double degree between ENSIC and UIS, with projects taken from simulation to real assembly.

Skills

- Mass and energy balances
- Equipment sizing
- Risk analysis (MOSAR, P&ID)
- Techno-economic assessment
- Experimental work and project management

Software

Aspen HYSYS / PRO II ★★★★★
Aspen Plus ★★★★★
Python / MATLAB / Excel ★★★★★
Fusion 360 ★★★★★
Claude / Claude Code ★★★★★

Education

- 2025 · present
Diplôme d'ingénieur + Master · Chemical and Process Engineering
ENSIC, Nancy, France. 3rd year. Specialisation: Processes for Energy and the Environment.
- 2020 · present
Chemical Engineering
Universidad Industrial de Santander, Colombia. 5th year.